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WBF Foreword

The purpose of this series of books from WBF, The Organization for Production Technology, is to publish papers that were given at WBF conferences so that a wider audience may benefit from them.

The chapters in this series are based on projects that have used worldwide standards—especially ISA 88 and 95—to reduce product variability, increase production throughput, reduce operator errors, and simplify automation projects. In this series, you will find the best practices for design, implementation, and operation and the pitfalls to avoid. The chapters cover large and small projects in a wide variety of industries.

The chapters are a collection of many of the best papers presented at the North American and European WBF conferences. They are selected from hundreds of papers that have been presented since 2003. They contain information that is relevant to manufacturing companies that are trying to improve their productivity and remain competitive in the now highly competitive world markets. Companies that have applied these lessons have learned the value of training their technical staff in relevant ISA standards, and this series provides a valuable addition to that training.

The World Batch Forum was created in 1993 as a way to start the public education process for the ISA 88 batch control standard. The first forum was held in Phoenix, Arizona in March of 1994. The next few years saw growth and the ability to support the annual conference sessions with sponsors and fees.

The real benefit of these conference sessions was the opportunity to network and talk about or around problems shared by others. Papers presented at the conferences were reviewed for original technical content and lack of commercialism. Members could not leave without learning something new, possibly from a field thought to be unrelated to their work. This series is the opportunity for anyone unable to attend the conferences to participate in the information-sharing network and learn from the experiences of others.

ISA 88 was finally published in 1995 as *ISA-88.01-1995 Batch Control Part 1: Models and Terminology*. That same year, partially due to discussions at the WBF conference, ISA chartered ISA 95 to counter the idea that business people should be able to give commands to manufacturing equipment. The concern was that business

people had no training in the safe operation of the equipment, so boardroom control of a plant's fuel oil valve was really not a good idea. There were enough CEOs smitten with the idea of "lights-out" factories to make a firewall between business and manufacturing necessary. At the time, there was a gap between business computers and the computers that had infiltrated manufacturing control systems. There was no standard for communication, so ISA 95 set out to fill that need.

As ISA 95 began to firm up, interest in ISA 88 began to wane. Batch control vendors made large investments in designing control systems that incorporated the models, terminology, and practices set forth in ISA 88.01 and were ready to move on. ISA 95 had the attention of vendors and users at high levels (project-funding levels), so the World Batch Forum began de-emphasizing batch control and emphasizing manufacturing automation capabilities in general. This was the beginning of the transformation of WBF into "The Organization for Production Technology." Production technology includes batch control.

The WBF logo included the letters "WBF" on a map of the world, and since this well-known image was trademarked, the organization dropped the small words "World Batch Forum" entirely from the logo after the 2004 conference in Europe. WBF is no longer an acronym. Conferences continued annually until the economic crash of 2008. There was no conference in 2009 because many companies, including WBF, were conserving their resources.

WBF remained active and solvent despite the recession, so a successful conference was held in 2010 using facilities at the University of Texas in Austin. Several papers spoke of the need for procedural control for continuous and discrete processes. The formation of a new ISA standards committee (ISA 106) to address this need was announced as well. Batch control is not normally associated with such processes, but ISA 88 has a large section on the design of procedural control. There is a need for a way to apply that knowledge to continuous and discrete processes, and some of those discussions will no doubt be held at WBF conferences, especially if the economy recovers. We would like to invite you to attend our conference and participate in those discussions.

WBF has always been an organization with an interest in production technologies beyond batch processing, even when it was officially "World Batch Forum." Over the years, as user interests changed, so has WBF. We have not lost our focus on batch; we have widened our view to include other related technologies such as procedural automation. We hope you will find these volumes useful and applicable to your needs, whatever type of process you have, and if you would like more information about WBF, we are only a simple click away at http://www.wbf.org.

William D. Wray, Chairman, WBF
Dennis L. Brandl, Program Chair, WBF
August 2010

Foreword by Walt Boyes

Many years ago, some dedicated visionaries realized that procedure-controlled automation would be able to codify and regularize the principles of batch processing. They set out on a journey that eventually arrived at the publication of the batch standard ISA 88 and the development of the manufacturing language standard ISA 95.

Many end users have benefited from the work of these visionaries, who founded not only the ISA 88 Standard Committee but also the WBF. WBF has been an unsung hero in the conversion of manufacturing- to standards-based systems.

Today, WBF continues as the voice of procedure-controlled automation in the process and hybrid and batch processing industries. The chapters that make up this book series provide a clear indication of the power and knowledge of the members of WBF.

I have been proud to be associated with this group of visionaries for many years. *Control* magazine and ControlGlobal.com are and will continue to be supporters of WBF and its aims and activities.

I would like to invite you to come and participate in WBF, both online and at the WBF conferences in North America and Europe that are held annually. You will be glad you did. You can get more information at http://www.wbf.org.

Walt Boyes, ISA Fellow
Editor in Chief
Control magazine and ControlGlobal.com

Preface

The chapters in this book are written by people who have implemented systems with the aid of various parts of the ISA-95 standard. There is very little untested theory here. The chapters are divided into three sections:

- Chapters 1 to 3 are concerned with ISA-95 in general.
- Chapters 4 to 13 discuss ISA-95 and Enterprise Resource Planning (ERP)–Manufacturing Execution System (MES) integration
- Chapters 14 to 20 are about ISA-95 and MES

Almost all the chapters refer to MES because the business systems (ERPs) do not yet speak the control system's language, starting with transactions. Some chapters refer to SAP, a multinational business software company. WBF normally discourages commercial references, but SAP was the 800-pound gorilla in the room and needed to be mentioned. SAP claims 102,500 customers according to its Web site (http://www.sap.com/usa/index.epx).

Charlie Gifford's excellent Chapter 1 is an overview of the problem of communication between the shop floor and the transactional business machines. To fix this, it introduces thirty-eight acronyms used in ten times as many places. To help avoid confusion, I've made an effort to expand all acronyms that were not expanded in the original paper.

Scott Sommers's Chapter 8 supplies a good introduction to the five Levels of ISA-95. Dr. Ted Williams of Purdue University developed the Purdue Reference Model for enterprise manufacturing in the eighties, which gave us the Levels that put MES in Level 3.

Manufacturing Operations Management (MOM) models are required to make sense of MES. The introduction to them can be found in Jean-Luc Delcuvellerie's Chapter 16.

Bianca Scholten's Chapter 17 has a great story about interaction with customers, leading to a complete outline for writing a User Requirements Specification (URS) document for any ISA-95 integration project. The outline is in the chapter's appendix and really deserves special notice.

WBF's Business To Manufacturing Markup Language (B2MML) is essential for communication among MES and ERP systems. Chapters 4, 7, 9, and 11 discuss the uses of B2MML and provide examples of code.

A Note on Style

The WBF series has been edited using the 15th edition of the *Chicago Manual of Style* (CMS). This is the reason that you will find that your favorite capitalized words are not capitalized in certain instances. As a community of people that read books and papers on manufacturing process automation and business-shopfloor integration, we are used to every other word being capitalized because it is the name of a concept or a program variable, schema, and so on. The capitalization of terms in specification documents only makes it worse.

The problem is mitigated by using acronyms that are defined once in each chapter. Some words and phrases that are capitalized are important concepts that should be differentiated from common usage; others really are proper names. The rest contain the desired information whether they are capitalized or not, and so the CMS prevails.

Historical Perspective

Two kinds of computer control systems have evolved since technology made them possible. The first types of systems that automated business procedures such as payroll, manufacturing profit and loss, human resources, and others were all derived from a monetary viewpoint. The second types of systems automated process control, beginning with the basic Proportional-Integral-Derivative (PID) loop and adding control functions as the computer and glass control panel became accepted.

Some business computer vendors tried to get into the process control field and failed for lack of knowledge of what they were getting into. Process control is concerned with maintaining and recording process measurements, not profit and loss. Operators want to know how the process is doing and want to have handles that will let them take corrective action, with less than a second elapsing between command and response.

And so it was that a great divide opened up between manufacturing control and business IT. Manufacturing requires uninterruptible computing power, with service required all day, every day of the week. Everyone in manufacturing has a story about a failure at 3 a.m. Business machines operated weekdays, and users

didn't mind having a few days off while the system was maintained. "Oh, I'm sorry but the computer is down right now. Can you call back tomorrow?"

This situation began changing as computer operating systems and communication systems settled down into something resembling standard operation. Businesses knew they needed information from Manufacturing because they'd been getting paper status reports, perhaps weekly.

All this was in flux when ISA95 was born in 1995. The original impetus came from the extravagant claims of MES vendors saying they could connect the boardroom to the shop floor. It is not to their credit that they did not understand that people in upper management had no training in the safe operation of their manufacturing facilities. There was no control knob for "profit" that the CEO could crank to its extreme position.

There is still a lot of talk about the shop floor. When the day comes that there is nothing more to do on the shop floor, marketing will discover that there is a basement under the shop floor and that it has data. Don't ask what data—they'll figure it out as they go. That will set off another round of product differentiators and books like this one.

Distributed Control Systems (DCSs) use parallel processing to crunch incomprehensible amounts of data. PID controllers may save some data from the last one or two iterations, but they are mainly concerned with current measurements and previous outputs. Historians may be able to capture the results of each cycle, but this fire hose stream of data needs to be condensed before a human can make use of it. Before computers, data were recorded on strip charts that moved at 3/4 of an inch per hour, making anything less than a minute or two lost in the width of the ink trace.

ISA95 set out to model the communication between business systems and the Manufacturing Control Systems (MCSs). First you have to model what's out there on the shop floor, then decide what's worth remembering, then organize it by manufacturing function, and finally you need two-way communication using a formal language between manufacturing and business. You can't design a language without knowing what has to be communicated, hence the need to make data models of manufacturing and business activities.

Communication means secure transactions to business people. Control people are more used to the publish-and-subscribe model because there is not enough time for secure transactions. Subscribers have ways of knowing when publication stops, just as you know when your newspaper wasn't on the doorstep. See Chris Monchinski's excellent Chapter 4 for a description of the use of publish and subscribe in business, as well as using ISA-95 principles when a full MES layer is not justified.

One other aspect of communication is not discussed in these chapters, and that is time synchronization among systems. Some systems are islands in time,

updated occasionally by somebody's wristwatch. An increasing number of systems use network time applications, like Simple Network Time Protocol (SNTP), to synchronize machines to Global Positioning System (GPS) time. GPS time receivers can be used to synchronize networks that are isolated by a data diode.

Considering all the messages that have to be time stamped, this is a good thing. The U.S. Food and Drug Administration (FDA) now requires time stamping as close to the source as possible. Foundation Fieldbus was designed to provide this capability in the early nineties. The limit of time resolution is set by the execution cycle of the control function blocks. A block must execute to get a value or announce an alarm, and that execution time is set by the macrocycle for block execution. Nobody executes all of them at once.

Defining a language requires syntax, vocabulary, and grammar that can be understood by both sides. Furthermore, computer networks require addresses and rules for packing a message into a bag of bits and also require ways to handle errors. Transmission Control Protocol (TCP) is a secure protocol because it uses handshake messages ("Did you get it?" "Yes, I got it") to establish communication for every bag of bits sent. TCP has rules for finding lost messages and recovering them by retries. The principle language for MES communication is now WBF's markup language B2MML, which is also used for Manufacturing To Business (M2B).

Markup languages began at IBM in the seventies with Generalized Markup Language (GML), invented by Goldfarb, Mosher, and Laurie. Later GML became Standard Generalized Markup Language (SGML). The first major application was Hypertext Markup Language (HTML), which allows zillions of computers to talk to each other using Web browsers. HTML evolved at a rapid pace, particularly as a result of conflicts between Netscape and Microsoft, and has now become "polluted" by all the stuff added for better marketing, tracking, and advertising.

Extensible Markup Language (XML) was privately developed by Bosak, Bray, Clark, and others in the nineties to restore purity. The World Wide Web Consortium (W3C) then developed Extensible Hypertext Markup Language (XHTML). Sun Microsystems introduced Java at about the same time, which was picked up by IBM and Oracle for communication among their many operating systems. Microsoft chose XML and wove it into the .NET system, mostly because it wasn't controlled by Sun. Now IBM has adopted XML, and so it goes.

Markup languages contain elements and attributes that both ends of the communication link must understand. An XML schema is a document that defines the elements and attributes, limits on the way they may be structured, their syntax, and the data types that may be used in a message. B2MML is a schema for the XML documents used for communication on a network of mixed MES and business system computers. Of course, XML is not used for any of the traffic on a

control network because it is too verbose, but there may be interface or gateway computers that use it.

Eventually, the Wild West that was MES adopted ISA95's work to home in on a standard way of selecting manufacturing data and exchanging it with ERP systems. That work is still evolving, just as HTML is evolving to fit the needs of users of the World Wide Web.

Bill Hawkins
August 2010

ISA-95 Business Case Evolves through Applications and Methodologies

Presented at the WBF North American Conference, March 5–8, 2006, by

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Abstract

Business To Manufacturing (B2M) data exchange applications and system life-cycle methods are being developed from the ANSI/ISA-95 Enterprise-Control System Integration Standard to adapt and optimize manufacturing in the 21st century "pull" marketplace. The MESA/ISA-95 Best Practices Working Group will publish an annual ISA Technical Report to document these evolving applications and methods with an explanation of the ISA-95 business case. The business case centers on the following:

- Lowering life-cycle cost of B2M interfaces and manufacturing operations applications
- Constructing the flexible Manufacturing Application Framework (MAF) to optimize B2M interoperability and production (capability) flexibility through B2M functional segregation for optimized production workflow

The standard practically addresses today's B2M language (terminology and schema) requirement for an Application To Framework (A2F) data exchange using Service-Oriented Architectures (SOA). Utilizing ISA-95, schema foundation, applications, methods, and business cases are established through a structured MAF consisting of (1) workflow function organization, (2) transformation best practices for operations applications, and (3) their transactional interfaces. The business case for ISA-95 is demonstrated by the fact that ISA-95 does the following:

- Enables application of Lean practices of Standard Work and value streaming
- Enables the development of B2M functional segregation methods to correctly position operational tasks within Enterprise Resource Planning (ERP), Supply Chain Management (SCM), and Manufacturing Execution Systems (MESs) to optimize single-piece workflow and supply chain flexibility
- Structures a life-cycle management framework to lower a system's Total Cost of Ownership (TCO) and execute MAF flexibility in response to market change

The MAF utilizes ISA-95 part 3, "Models of Manufacturing Operations Management," to explain the influence of different system configurations on workflow, life-cycle cost, flexibility, and change management.

Recognizing a Historical Inflection Point in World Industry and Markets

Beginning in 1995, the Instrumentation, Systems, and Automation Society (ISA) and WBF developed the ANSI/ISA-95 Enterprise-Control System Integration Standard and the Business To Manufacturing Markup Language (B2MML). These works are intended as the foundation for standardized best practices for information exchange between plant systems and plant-to-business systems. Over the

last 10 years, Manufacturing Operations Management (MOM) solutions evolved to enable the distributed supply chain networks for 21st-century markets. ISA-95 based MOM applications and methods are recognized as the foundation for configurable, interoperable software tools to integrate interoperable data in readily useful forms to extended enterprise systems. This chapter assumes that the reader is familiar with ISA-95 and B2MML and so will focus on best practices and business cases and will not provide a standards overview.

Goal of the "ISA95/MESA Best Practices Technical Report"

The annual "ISA95/MESA Best Practices Technical Report" will explain how the ISA-95 Enterprise-Control System Integration Standard is applied to lower TCO of MOM systems and their enterprise and plant interfaces. ISA-95 best practices propose a three-legged MAF containing the following:

1. *Tools*. ISA-95 methods and technical applications that characterize, support, and adapt production workflow processes
2. *Training and staffing*. Defined system roles and skill sets for personnel for MOM processes
3. *Delivery*. A defined transformation and life-cycle management process for MOM

The ultimate goal of the ISA-95/MESA Best Practices Working Group is to explain "how to" apply, migrate to, and maintain a single data definition across Level 3 functions and interfaces, the MOM domain, and their Level 4 domain enterprise interfaces. By utilizing developing ISA-95 methodology and technical applications, the TCO for manufacturing IT architectures as well as manufacturing and supply chain operational costs are dramatically reduced. The annual "ISA-95 Best Practices Technical Report" consists of a series of related "how to" white papers described in the context of ISA-95 models, definitions, data flows, and the Level 3 interfaces between Level 4 enterprise functions and Level 2 shop floor systems.

Manufacturing Trends Relevant to the Role of ISA-95

The 21st-century manufacturing model is all about flexibility of production capabilities within globally distributed supply chain networks or Demand-Driven Supply Networks (DDSNs; as described by AMR Research). Manufacturing markets are

rapidly changing and driven by global competitive trends that make production flexibility a critical path component of supply chain collaboration. This coordinated data exchange across global supply chains and internal enterprise groups is just a part of the ISA-95 business case. Current industry discussions are focused on production's actual role in e-commerce, product development, supply chain planning and replenishment, or logistics. For any 21st-century manufacturer to be competitive, actual manufacturing operations activities must be highly interactive in supply chain and enterprise processes for effective collaboration and competition. This is the domain of collaborative and flexible MOM system architectures. This chapter explains the business cases for using evolving ISA-95 methods to effectively design, implement, change, and optimize the MOM business processes and for supporting MOM system architectures within the larger DDSN model.

Global Business Drivers for Flexible Manufacturing

Each vertical industry is being influenced by their unique combination of the following global business drivers for flexible manufacturing:

- Increased globalization: global markets with distributed sources of supply, production, and distribution facilities
- Increased customer diversity: culturally and geographically
- Increased access to competitive data
- Increased level of expected value
- Increased outsourcing of production and logistics operations
- Increased pace of new product introductions
- Increased product quality at lower cost

Today's competitive environment requires new business models that accommodate changing geographic presence, cost base, product array, use of new materials and technologies, and relationships with customers, suppliers, and other trading partners (Fig. 1.1).

The other side of the challenge includes organizational issues and aligning the goals and objectives of the different players in the organization as the company designs their 21st-century business model. Manufacturers that adopt a standardized approach to understanding, implementing, and deriving benefits from MOM applications should be able to accelerate the transformation to a successful implementation while increasing satisfaction levels of users.

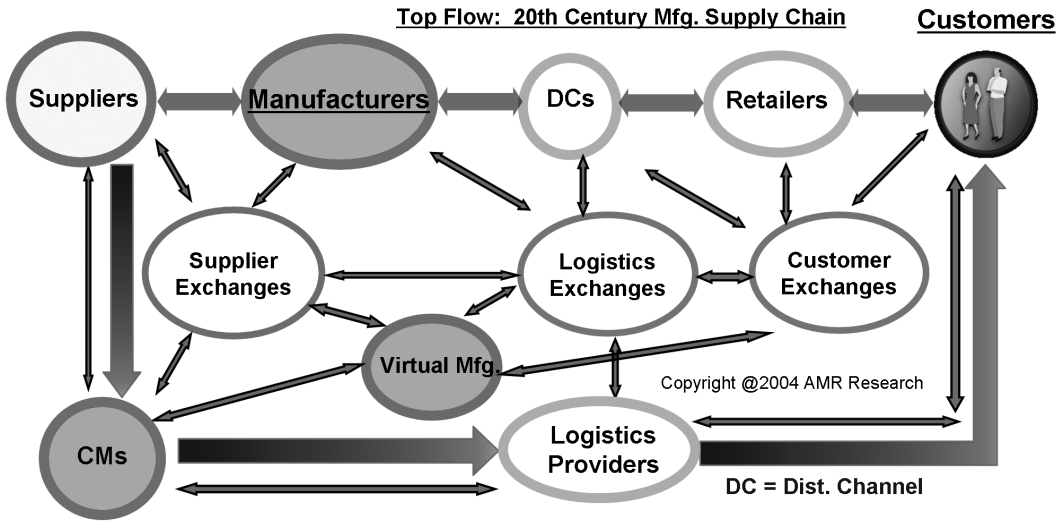


Figure 1.1. Aligning manufacturing capabilities to 21st-century challenges mandates change.

The B2M integration of operational systems requires unique skills to accelerate the B2M transformation. A thorough understanding of the following is essential:

- Operational processes and workflows
- Real-time controls and automation systems
- Information technology systems
- Knowledge of standards that encompass all levels of information flow
- Lean manufacturing and Six Sigma methods

The current MOM skills sets available in the marketplace are often segregated across these knowledge items by different levels; team members must become more broadly based to successfully integrate MOM architectures horizontally as well as vertically across the different levels.

DDSNs Create the Interactive 21st-Century Manufacturing Model

To effectively compete in 21st-century markets, companies are creating improved tangible value by accurately matching products and value-add services to each customer's need within hours rather than weeks. Timing is the key to profitability now. This value objective is not new, but maturing capabilities of Web technologies

(integrated into application software) have provided the interactive tools for the required collaborative and interoperable communication. In conjunction, ISA-95 is defining MOM data structures and exchanges (definitions) that can be utilized to construct the MOM Web services and SOA for MOM solutions. The operations management data and transaction definition in ISA-95 parts 3 to 6 are the basis of SOA for Manufacturing (SOAM) or a Manufacturing Services Architecture (MSA). These ISA-95 parts are establishing the real-time basis to quantify cycle time, cost, and resource elements of workflow for production, maintenance, inventory, and quality. As seen in Figure 1.5, B2M data flows and metrics are defined in terms of four B2M categories of information (activity definition, activity capability, activity schedule, and activity performance) for the four primary plant activity models shown. These data elements are required for scheduling and planning order fulfillment across DDSNs. In 2005, ISA95 and the Supply Chain Council formed the ISA95/SCOR Alignment Working Group, which mapped data flows between the Supply Chain Operations Reference (SCOR) and ISA-95 part 3 models. This is the foundation for aligning the development of the two standards, as ISA-95 parts 4 and 6 are composed over the next few years. This combined work enables the rapid evolution of MSA and DDSN architectures.

SOA Components

Over the last 15 years, integration technology has evolved from data to process level capabilities with SOAs being the merger of business process management and the enterprise services bus. SOA surrounds Web services containing the business process rules with various technologies to manage, orchestrate, and choreograph Web services into an executable business model. Core SOA services include the following:

- Services registry
- Enterprise services bus
- Web services management
- Web services security and identity management
- Web services development and programming tools

The business justification for ISA-95 based MSA within interactive MOM solutions is reinforced by contract manufacturers. The necessary practice of outsourcing production to contract manufacturers is driven by global markets in growing economies such as India, China, and Eastern Europe, across all industries. The result is low-cost competition for North American and European suppliers, due to lower labor and operating costs. Also, outsourcing has accelerated adoption

of the DDSNs model where consumer demand for a single product order is now met by evaluating competing supply chain paths. Basically, the order fulfillment path is now determined by evaluating real-time supply chain cost to customer demand for on-time delivery at a specific quality level. Order commitments are made based on this algorithm (Fig. 1.2).

Original Equipment Manufacturers (OEMs), such as IBM, GE, and HP, previously known for building a variety of products, are now known for their market-leading product designs and their ability to market and sell them by managing their DDSNs through contract-manufacturer partners and internal production. MSA-based MOM solutions provide the means for OEMs to identify available materials and resources (capacity) across competing supply chains in a lead time versus price form for immediate prototyping to market demand (i.e., Design For Supply [DFS]). The more real time the SCM, the larger the profit margin due to order accuracy.

As OEMs evolve their MSA practices, they build a tighter relationship with key suppliers and contract manufacturers to (1) provide forecasted demand from all customers and sales channels and (2) require real-time production records and visibility to the OEM customer from ISA-95 based MOM solutions (Fig. 1.3).

To address global competition, 21st-century manufacturers are rapidly adopting several types of corporate software systems to transform to their global business model: ERP, SCM, Supply Chain Execution (SCE), Customer Relationship

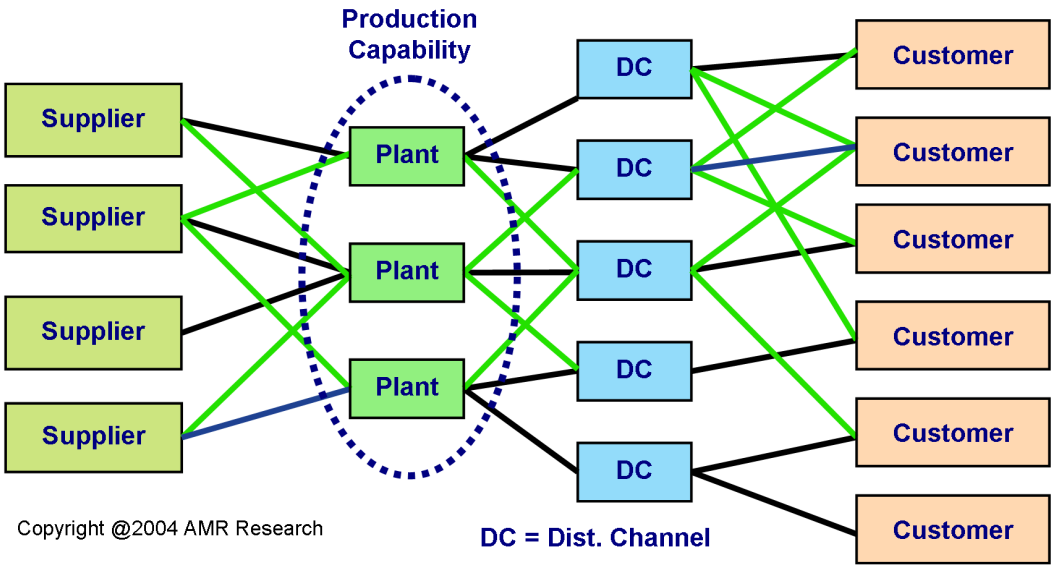


Figure 1.2. Business evolves into configurable DDSN.

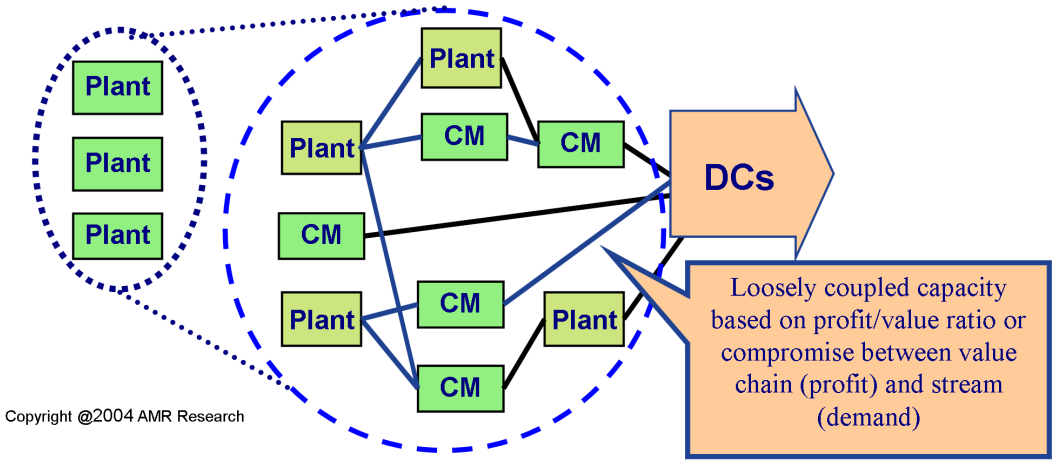


Figure 1.3. Production capability configured in real time: Evaluate customer value and On-Time Delivery (OTD) to production path and profit margin.

Management (CRM), design collaboration tools, Product Life-cycle Management (PLM), and others. These extended enterprise systems were supposed to be designed to exchange information outward to customers and suppliers in near real time. Results over the last 10 years have been poor to fair, since these tools were not designed for a DDSN global market. For some companies, these early generation systems provided quick benefit by reducing the time and costs of interacting with their 20th-century linear supply chain partners; however most have not achieved the predicted benefit due to lack of actual production data integrity (response and accuracy). An ISA-95 methodology for data exchange is being developed for the next-generation ERP system to support the complex (nonlinear) global DDSN model. This methodology will help map data and transactions between workflows of a distributed supply chain and production. With inaccurate or "too coarse" production data being the major limitation of the Y2K corporate business systems, a similar major issue is a lack of commonly defined metrics or schema across supply chain and production operations. ISA-95 methods solve these limitations for system interoperability. These will be further explained as part of the ISA-95 business justification.

ISA-95 Blends the SOA Approach into MOM

The ISA-95 standards address the interface or exchange of data between the extended enterprise systems (sales, planning, scheduling, and procurement) and the following part 3 models of MOM (and example MOM systems):

- *Production Management Operations*. Product tracking and tracing, manufacturing execution, manufacturing intelligence portals, finite capacity and detailed scheduling, work order management, production sequencing, batch execution, recipe management, and so on
- *Maintenance Management Operations*. Asset management; computerized maintenance management; preventive maintenance; Maintenance, Repair, and Operations (MRO); and so on
- *Quality Test Management Operations*. Statistical process control, statistical quality control, laboratory information management, Corrective And Preventive Action (CAPA), Material Review Board (MRB), and so on
- *Inventory Management Operations*. Management Resource Planning (MRP), tracking of plant-side raw material, Work In Process (WIP), finished goods, and so on

The following lists the ISA-95 foundation for MSA, that is or will be contained in parts 3 through 6:

- *ANSI/ISA-95.00.03-2005 Part 3: Models of MOM*
- *ISA-95.00.04 (Draft) Part 4: Object Models and Attributes of MOM*
- *ANSI/ISA-95.00.05-2007 Part 5: Business to Manufacturing Transactions*
- *ISA-95.00.06 (Proposed) Part 6: MOM Transactions*

With this work, the MSA concept brings the previously mentioned MOM operations of a manufacturing business into alignment to intelligently respond to market forces. Currently, corporations are attempting to identify MSA practices to publish and distribute customer demands across the supply chain. Once all suppliers in company's DDSN are able to align on market demand, products are then rapidly and accurately developed for new markets while maintaining high margins. The combination of a DDSN model being driven by an ISA-95 MAF allows a manufacturer to capture a large market share or even create markets, due to their ability to rapidly adapt their collaborative production resources to real-time market demand.

After Part 6, B2MML Is Required to Meet MSA Requirements and End-user Demand

B2MML version 4 is based on the following first approved, untested versions of ANSI/IEC/ISO/ISA-95 parts 1 and 2 in 2000 through 2002:

- 2000: *ANSI/ISA-95.00.01, Enterprise-Control System Integration, Part 1: Models and Terminology*
- 2002: *IEC/ISO 62264-1 Part 1: Models and Terminology*
- 2001: *ISA/ANSI-95.00.02 Part 2: Data Structures and Attributes*
- 2002: *IEC/ISO 62264-2 Part 2: Data Structures and Attributes*

In the current 5-year review of the first versions, many changes are being proposed due to end user lessons learned in applying the standards and schema applications (not to be addressed in this chapter). However, it is important to note that B2MML version 4 is a special case of manufacturing environments and the B2M interface instance, as opposed to a general case. This is illustrated in Figure 1.4 and Figure 1.5. B2MML version 4 is based on an academic definition at the Level 3 to Level 4 interface described in parts 1 and 2, where all MOM functions are plant side systems. B2MML schema has not yet evolved (and will not be able to evolve, since schema must follow the approved standard) to address a more wide range of real-world B2M interfaces where many MOM functions are within centralized corporate applications. With the release of part 3 and 5—and the eventual completion of parts 4 and 6—B2MML is evolving to adequately address Level 3 MOM data and workflows for a majority of hybrid manufacturing environments.

Most plants are hybrid environments from dock (raw materials) to dock (finished goods packaging) with a mix of work order types for customer orders (e.g., 80% Make To Stock [MTS], 10% Make To Order [MTO], 10% Engineer To Order [ETO]) and more of a mix of WIP work cell orders (50% MTS, 35% MTO, 15% ETO or rework). The B2M interface line ("B2M" line in Fig. 1.5) is determined by the MOM applications required at the plant floor to address the complexity of the work order mix and its associated workflow business rules. As evolving global markets drive manufacturers to rapidly adjust their work order mix based on market demand and drivers (i.e., toward a higher percentage of MTO and ETO), B2M functional segregation of MOM applications and the representative B2M interface line are determined by the MOM architecture that optimizes production, single-piece flow, profit margin, and throughput. Basically profit margin drives the MOM architecture. A manufacturer's ability to rapidly adapt their MOM architecture to new market conditions determines the level of success in their global markets. This proposed ISA-95 methodology is Step 3.5 in the MAF best practices methodology outlined later in the chapter.

The following is a list of the components of a simplified workflow complexity matrix:

1. Production types
   - Discrete manufacturing
   - Batch processing
   - Continuous processing

2. Work order types
   - ETO
   - MTO
   - MTS
   - Nine primary combinations with many hybrids (each have a specific set of business processes and rules)
   - Complexity contributors (e.g., product, legacy, speed, volume, color, size, compliance, SKU count)

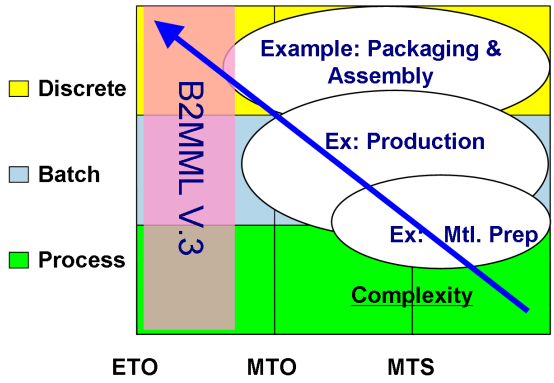


Figure 1.4. Simplified workflow complexity matrix.

Once industry has agreed upon the standard form of B2MML, the ISA-95 body of work (schema, standards, applications, and methods) will adequately model the majority of the Level 3 MOM use cases, data flows, transactions, business processes, and metrics (interface, Key Performance Indicators [KPI], and operational) construction. Based on this MOM use case modeling, vendors will roll out their collaborative libraries of MSA Web services that end users essentially need for global DDSN architectures. At this point (2008–2010), the ISA-95 methodology

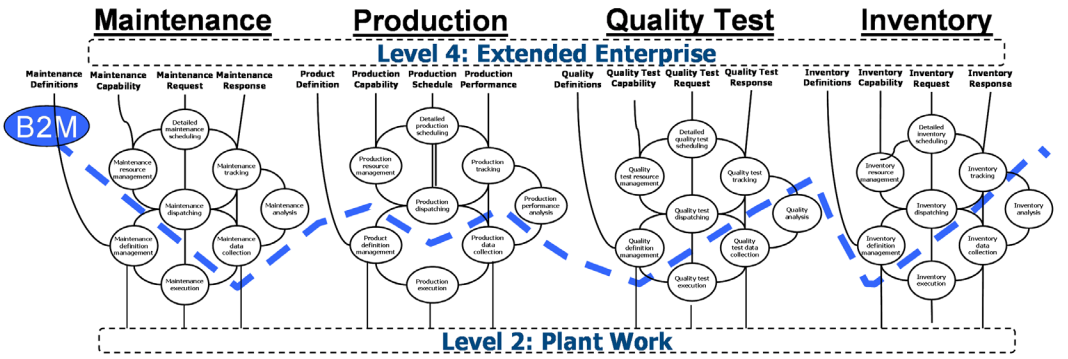


Figure 1.5. ISA-95 part 3 hypothetical B2M interface chart: Real-world B2M interfaces exist within MOM activity model functions.

for construction of the MAF will coalesce and mature into a proven SCM system required for the DDSN implementation. This chapter and the ISA-95 Technical Report (First Edition 2006) proposes the working methodologies to drive toward these goals over the next few years. Software vendors and MOM literature are moving to this important inflection point for the next 3 to 5 years to meet the 21st-century manufacturing model requirement. As of 2006, the ISA-95 body of work needs accelerating to limit the scale and variation of implemented beta B2MML interfaces and MOM applications. Many early innovative adopters are struggling with how to apply and extend B2MML version 4 to address their hybrid environments and are taking their best guess as to where the ISA-95 body of work will direct software vendors in the future.

Industry collaboration is the real-world challenge in order to lower TCO for integrated systems through focused, dedicated effort. Accordingly, the end user's commitment to B2M interoperability will remain speculative until this open standard's work has been organized and accomplished. Industry analysts, vendors, and end users are all looking to each other for leadership for this MOM standards effort. No clear leader has yet emerged as of this writing. A loosely coupled group of end users, vendors, and consultants are proactively forming alignment working groups, but the progress is much slower than the market need.

The ISA-95 Business Value: Low Cost B2M Interfaces and Flexible MAF

Manufacturing data exchange applications and system life-cycle methods are being developed from ISA-95 standards. ISA-95 applications and methods are being globally applied by innovative manufacturers to adapt and optimize manufacturing for the 21st-century "pull" markets.

The ISA-95 business case is centered on the following:

- Lowering the life-cycle cost of B2M interfaces
- Constructing the flexible MAF to optimize B2M interoperability and production (capability) flexibility through B2M functional segregation for optimized production workflow

The standard practically addresses the language (terminology and schema) requirement between operations and business systems for an Application To Framework (A2F) data exchange required for MSA.

The business case is further established by providing organization and best practices to transform operations applications and their transactional interfaces

into a flexible manufacturing framework. Through its B2M functional and object models and single Extensible Markup Language (XML) schema hierarchy for B2M integration, ISA-95 transformation methods merge production workflows into the overall collaborative business process. The merger of ISA-95 models, applications, and methods to form the MAF is proposed by the ISA-95/MESA Best Practices Working Group in the annual ISA Technical Report. MAF defines data exchanges and metrics for integrating the production systems (1) horizontally between MOM applications and (2) vertically between global DDSN and enterprise systems.

ISA-95 business value and MAF are derived from two classes of best practices to illustrate the high value of the MOM application development and life-cycle processes utilizing ISA-95. Both classes of best practices are early in their life cycle and are rapidly advancing due to endorsement by the end user and vendor community such as SAP and Microsoft. These ISA-95 best practices are being applied in isolated MOM projects by the most innovative manufacturers in the world, such as Procter & Gamble, Dow, Arla Foods, Nestlé, DuPont, and BP. Widespread use has not occurred due to the immature state of MOM solutions and SOA manufacturing technologies.

Class 1: ISA-95 Technical Applications for Improving B2M Interface Interoperability

The ISA-95 technical applications utilize the following:

- B2MML schemas
- ISA-95 models from parts 1 to 3
- Proposed part 4 Level 3 MOM data flows and operations metrics
- Part 5 B2M transactions
- Proposed part 6 MOM transactions
- Other developing information technologies such as XML, Web services, SOAs, or data exchange frameworks for application interoperability

Class 2: ISA-95 Transformation Methodology for Improving B2M Operations Interoperability

Class 2 describes a set of operations transformation methodologies that align with other current best practices for the software development life cycle such as Good Automation Manufacturing Practices (GAMP) or Microsoft Enterprise

Framework (MEF) methods. These ISA-95 based methods are intended to be used to construct an MAF for optimizing B2M operations interoperability by providing MOM system architectures that are able to adapt to market changes in the company's DDSNs.

This chapter simply outlines and briefly describes the "ISA-95 Best Practices Classes." This white paper introduces the evolving best practices and explains the high-level business case through market need. In future white papers from the ISA95 Best Practices Workgroup, a business justification for each application and methods will be explained using the following five steps:

1. List the proposed ISA-95 best practice
2. Identify the prioritized business driver and operational benefits
3. Identify the current state and the underlying forces (reasons for underperformance, stakeholders, resources, etc.) behind the key driver (using quantification examples if possible)
4. Analyze the example of capital expenditures, recurring costs, and recurring savings
5. Provide an example of a net present value analysis

This writing utilizes a Six Sigma structure to explain the construction of an MAF using Class 1 and 2 best practices. The Six Sigma structure is an abstraction of the following Define, Measure, Analyze, Improve, and Control (DMAIC) process:

1. *Define*. Determine project objectives, scope, resources, and constraints.
2. *Measure*. Determine Critical-To-Quality (CTQ) tasks to production workflow. Obtain data to quantify process performance.
3. *Analyze*. Analyze data to identify root causes of production workflow disruptions and defects.
4. *Improve*. Intervene and change current MOM processes to improve performance through the ISA-95 transformation to a single schema across MOM systems.
5. *Control*. Implement an MAF or life-cycle management framework to maintain workflow performance through analysis of market to production workflow and system architecture.

In applying Class 1 best practices of ISA-95, "Technical Applications to Improve B2M Interface Interoperability" white papers utilize ISA-95 models to

define best practice. These technical applications are the foundation for B2M and MOM interface interoperability by providing the data hierarchy and definition for interface construction. The ISA-95 MAF is then able to be constructed through Class 2 methodologies.

Step 1: Train Staff and Benchmark Technical Applications to Improve B2M Interface Interoperability (Define)

Training and benchmarks should include the following:

- B2M functional model
- B2M interface object model and attributes
- B2MML
- Part 3 MOM data flows mapped between SCOR data flows
- Proposed part 4 Level 3 MOM data flows and operations metrics
- Part 5 B2M transactions
- Proposed part 6 MOM transactions

In applying the Class 2 best practices of ISA-95, "Transformation Methodologies to Improve B2M Operations Interoperability" white papers explain the step-by-step process to improve manufacturing operations interoperability through the adoption of a single schema system architecture that supports development of an MAF for system life-cycle management for 21st-century markets.

Step 2: Structured Manufacturing Operations Assessment and Schema Migration Plan (Define, Measure, Analyze)

In order to map out the manufacturing transformation, a manufacturing operations assessment and strategy should be prepared to characterize a company's cultural and technical maturity, risks, and system road map, with a business justification for each project. The following are the minimal components of a manufacturing operations assessment:

- "As is" MOM assessment
- "To be" MOM gap analysis
- Prioritized criteria for business cases for MOM applications

Typically, an organization will not transform to ISA-95 based single schema MOM architecture in a short period of time. There is an established set of disparate terminologies, workflows, data flows, and applications that are utilized in running ongoing manufacturing operations and supply chain processes. A migration plan will involve a pre-migration step.

A good pre-migration step includes a study of operational and business drivers for a potential transformation. Since MOM covers a wide range of functions at the manufacturing level, the business drivers prioritized with regard to quickest and highest returns may assist in identifying the following information:

- MOM-related business drivers
- Priority of MOM-related business drivers
- Returns based on a net present value analysis
- Highest probability of success based on returns and current needs

The step described previously, "Prioritizes criteria for business cases for MOM applications," is required to assess the magnitude of investment and effort required in the transformation process. A sound approach includes an assessment of current MOM elements combined with a gap analysis derived from a comparison of the current MOM elements to the ISA-95 based MOM elements.

Step 3: Accelerated MOM Transformation and Application Framework Implementation (Analyze, Improve, Control)

To accelerate a manufacturer's transformation to an MAF, the following projects should be undertaken:

- Establish a project plan standard for MOM systems
- Establish standard design criteria for MOM knowledge management
- Establish schema standards for enterprise and MOM systems
- Build MOM flexible MAF
- Analyze functional segregation between enterprise and MOM applications
- Simplify functional requirement process for MOM functions and interfaces
- Simplify extended enterprise metric construction process
- Simplify event management construction process

Step 4: Life-cycle Management of MOM Application and Interfaces (Analyze, Improve, Control)

To create a sustained operations culture for continous improvement, the manufacturer should restructure it's organization to have a Manufacturing-System-Technology (MST) group to own, govern, and continuously improve the MAF. The following are tasks for this MST group:

- Use the flexible manufacturing (application) framework for change management planning and the change management process
- Simplify the workflow analytic development process
- Simplify the quality analytic development process
- Simplify skill-set mapping of super users, system and data owners, functional users, and process modelers

The biggest challenge faced by various systems and engineering departments is ensuring that the implemented MOM systems are well accepted and utilized by the user community within a manufacturing operation. An effective change management plan with process expeditors developed as part of the best practices approach alleviates some of the risks associated with nonacceptance. Educating the user community and stakeholders about the benefits of MOM applications is an ongoing, challenging cultural issue. ISA-95 MOM architectures require the collection of large amounts of data from automated as well as manual sources. Manual sources of data are typically where questionable acceptance plays a significant role in application success. User acceptance depends on a well-defined workflow combined with ongoing validation of the benefits to individual user groups and stake holders.

Manufacturing companies typically implement point solution MOM systems that are absolutely essential for short-run requirements where the politics force a bypass of some elements of ISA-95 best practice methodologies in order to ensure rapid deployment of the system. Table 1.1 highlights some differences in taking a short-term project approach with the best practices. As noted earlier, the best approach may be to follow the best practices model with elimination of a few steps in order to achieve the rapid deployment for short-term goals. Deviating from the best practice at the local project level may create a risk of nonstandardized approach but could be minimized if the deviation is kept to minimal levels.

Table 1.1. ISA-95 Business value: Holistic versus project approach		
Elements	*Generic best practices*	*Project-specific best practices*
Business drivers	Standard business and operational metrics (enterprise-wide as well as plant specific) Definition of metrics	Prioritize issues related to standard metrics and any localized metrics, based on urgency, need, returns, and so on.
Assessment standards	Function, activity, and task data flow based on ISA-95 models	As-Is MOM assessment Target versus current gap analysis
Implementation standards	Generic project plan Generic requirements process Standard design criteria for MOM ISA-95 based enterprise schema standards Common components Boundaries between different levels	Application-specific project plan Application-specific requirements Application-specific schema Common and application-specific components Interfaces based on boundaries
Life-cycle management standards	Changed management planning and process via flexible MAF Simplified workflow analytic development process Simplified quality analytic development process Skill-set mapping of super users, owners, users, and process modelers	Application-specific maintenance process Application validation of value delivered Application-specific change management process

ISA-95 MAF Enables a 21st-Century Lean Manufacturing Renaissance

MSA best practices will provide companies with supply chain data and software technology to apply Lean manufacturing by balancing profit, quality, and cost against each other for the On-Time Delivery (OTD) commitment decision:

- The value chain (lowest cost path to customer) to drive maximum profits
- The value stream (value-added path to meet customer's expectation) to deliver quality products and services
- The cost of product throughout its life cycle

ISA-95 describes the basis for Standard Work, which is the foundation for Lean transformation and single-piece or order flow. ISA-95 functions, tasks, and data exchanges become the Standard Work component necessary to simplify and design an MSA for a Lean supply chain. MSA that are structured for single-piece flow interact with SOA processes across the distributed supply chain of the supplier and customers to build global DDSNs. MOM solutions utilizing MSA allow appropriate organizations to make better decisions using timed, event-driven, role-based data sets that are mapped into production workflow use cases characterized by ISA-95 models and B2MML. Currently, manufacturers utilize a wide range of Lean supply chain processes that only function correctly when accurate, real-time MOM information is available and accurately directed for MTO OTD supply chains:

- *Defined customer value stream.* Benchmarking and fine-tuning production activities and quality into DDSN
- *Just-In-Time (JIT) transportation and distribution*. Coordinating "pull" logistics
- *Value-added engineering and design*. Refining product characteristics to reduce waste
- *MTO sales and marketing*. Mapping customer specification and due dates into DDSN
- *Procurement*. JIT inventory levels for replenishment and fulfillment
- *Operations*. Proactively preventing equipment breakdowns with Total Productive Management (TPM) and Overall Equipment Effectiveness (OEE) methods

To optimize the 21st-century manufacturing enterprise, companies are recognizing that production workflow, use cases (transaction sequence), and data flows must be identified, characterized in an SOA, and optimized by utilizing Lean-manufacturing and/or Six-Sigma characterization techniques. As KPI and operations metrics—with their cause-and-effect relationships (compromises) to production workflow—are developed and built into systems, ISA-95 is the enabling tool for executing this functional design efficiently. Lean MOM applications transform previously optimized Lean workflows as global markets demand change.

Lean MOM Examples

Some examples of Lean MOM include the following:

- Standard Work single schema product tracking, genealogy, and performance reporting
- Finite capacity scheduling with single-piece flow
- Theory of Constraints (TOC) for line balancing
- Utilization management using Overall Equipment Effectiveness (OEE) with resource efficiency and benchmarking to drive cultural change
- Statistical Process Control (SPC; on-line, at line, off-line) and Laboratory Information Management Systems (LIMS) for quality and workflow statistical analysis
- Role-based manufacturing portals for interdepartmental communication of real-time situations with defined event management sequences (rules)

Conclusion

This 21st-century environment requires companies to evolve their 20th-century manufacturing business model, support systems, and existing organizational practices simply to survive. They need the ability to share data and information in a secure environment so that decisions are completed more rapidly and reliably, saving time and money. The current progress of MOM applications and methods is highlighted by the following:

- Public MOM standards and methodologies are being endorsed by end users and vendors.
- MOM solutions have become part of interactive global DDSN processes.
- Vertical industry libraries of use cases and processes are being characterized using ISA-95. Resulting MOM software applications are more configurable and less a custom extension. Vendors are developing large libraries of use cases with configurable components, XML schemas, and templates toward their MSA framework for configurable interoperability.
- B2MML interfaces require much less custom interface development due to ISA-95 based libraries of configurable interfaces.

- The skill set required for MOM implementations is now recognized as a mixture of business process, IT, and manufacturing process skills.
- The return on investment for MOM solutions has been better quantified, explained, and accepted due to a large increase in repeatable MOM application sets at lower cost.
- Life-cycle costs for MOM systems are now more predictable due to an increase in tool functionality and a dramatic reduction in custom programming of interfaces and applications.

ISA-95 based MSA enables the flexible MAF of MOM systems to analyze and aggregate MOM data (e.g., capacity, capability, inventory, order and equipment scheduling) and then exchange data with Advanced Planning and Scheduling (APS), ERP, and SCM systems and the DDSNs. Developing ISA-95 best practices will provide consistency and flexibility to an extended enterprise by working interactively in real time within the supply chain to definitively determine the transformation rate to create new markets and move into them. This is 21st-century manufacturing. They enable decision making based on measurable and specific manufacturing constraints, abnormal conditions (e.g., alarms), and events. Flexible manufacturing is especially important as the United States, Europe, and the rest of the industrial world become more of a multilingual cultural melting pot. As mixing cultures, languages, foods, and fashion drive higher demand for niche and make-to-order products, manufacturers need responsive DDSNs and flexible plants to produce short, profitable production runs.

Acronym Glossary

A2F: Application To Framework
APS: Advanced Planning and Scheduling
B2M: Business To Manufacturing
B2MML: Business To Manufacturing Markup Language
CAPA: Corrective And Preventive Action
CRM: Customer Relationship Management
CTQ: Critical To Quality
DDSN: Demand-Driven Supply Networks
DFS: Design For Supply
DMAIC: Define, Measure, Analyze, Improve, Control
ERP: Enterprise Resource Planning
GAMP: Good Automation Manufacturing Practice

ISA: Instrumentation, Systems, and Automation Society
JIT: Just In Time
KPI: Key Performance Indicator
LIMS: Laboratory Information Management System
MAF: Manufacturing Application Framework
MEF: Microsoft Enterprise Framework
MES: Manufacturing Execution System
MESA: Manufacturing Enterprise Solutions Association
MOM: Manufacturing Operations Management
MRB: Material Review Board
MRO: Maintenance, Repair, and Operations
MRP: Management Resource Planning
MSA: Manufacturing Services Architecture
OEE: Overall Equipment Effectiveness
OEM: Original Equipment Manufacturers
OTD: On-Time Delivery
PLM: Product Life-cycle Management
SCE: Supply Chain Execution
SCM: Supply Chain Management
SCOR: Supply Chain Operations Reference
SOA: Service-Oriented Architectures
SOAM: SOA for Manufacturing
SPC: Statistical Process Control
TCO: Total Cost of Ownership
TOC: Theory Of Constraints
TPM: Total Productive Management
WIP: Work In Progress
XML: Extensible Markup Language